

Tingyu Wu

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RESEARCH INTERESTS

Stochastic optimization, decision-making under uncertainty, and statistical learning theory, with an emphasis on methodological development at the intersection of operations research, statistics, and machine learning.

EDUCATION

University of Illinois Urbana-Champaign

Aug. 2023 – Dec. 2026 (Expected)

B.S. Mathematics, Concentration in Data Optimization, Minor in Statistics

Selected Coursework: Real Variables, Nonlinear Programming, Linear Programming, Probability & Statistics II

RESEARCH EXPERIENCE

Undergraduate Research Assistant

Jul. 2025 – Present

Prof. Yifan Hu, Department of Statistics, Rutgers University

Remote

Stochastic NegDRO for Causal Invariance Learning

- Developed an online primal–dual method for NegDRO, leveraging environmental heterogeneity to establish finite-time expected squared-error bounds for time-averaged and averaged-iterate estimates.
- Formalized the conditional convergence proof in Lean 4 and Mathlib. [Research note](#) · [Lean code](#).

Sample-Average Approximation and Stochastic Minimax Optimization

- Studied near-stationarity guarantees for sample-average approximation.
- Applied the analysis to nonconvex–concave stochastic minimax problems to derive corresponding near-stationarity guarantees.

Causal Identification in BLP Demand Models (ongoing)

- Investigate gradient invariance across environments as an identification condition in BLP demand models, examining connections to instrumental-variable and GMM formulations.

Undergraduate Research Scholar

Sept. 2025 – Dec. 2025

Prof. Felix Leditzky, Illinois Mathematics Lab (IML), University of Illinois Urbana-Champaign

Champaign, IL

- Benchmarked classical optimization methods for entropy-based problems in quantum information theory, focusing on convergence and numerical stability.
- Analyzed algorithmic performance via performance and data profiles and implemented optimization routines for numerical evaluation.
- **Awarded the Nancy D. Anderson Undergraduate Research Award** for this project; [manuscript](#).

PROJECTS

Agent-based CBT Intervention | *LLM, LangChain, JSON, Agent, Document AI, Python*

May 2025 – Jul. 2025

- Built an intelligent agent to provide initial mental health screening and personalized CBT suggestions based on student questionnaire scores.
- Developed a complete pipeline including MHT score parsing, JSON conversion, and interactive agent responses.

LaTeX-OCR | *LaTeX, LLM, Im2LaTeX, Document AI, Python*

Nov. 2023 – May 2024

- Generated a robust dataset to train the LaTeX-OCR model using the 7B Zephyr LLM.
- Analyzed open-source LaTeX-OCR models, identified key distinctions, and implemented optimization strategies.

Hyper-parameter Optimization | *Neural Network, Genetic Algorithm, Transfer Learning*

May 2022 – Sept. 2022

- Collaborated on a group paper, [Hyper-parameter Optimization Based on GA & Transfer Learning](#).
- Compared existing hyper-parameter optimization methods and designed a novel algorithm leveraging genetic search and transfer learning.
- Implemented experiments on the MNIST dataset and demonstrated improvement.

SKILLS

Python, R, Lean, LaTeX, Git/GitHub